

1. Project Title

AI Ambulance Intelligence System: Machine Learning-Based Emergency Response Time Prediction and Decision Support

2. Comparative Analysis

S.No	Reference	Problem Addressed	Methodology/Technique	Dataset	Results/Performance	Limitations	Research Gap
1	Aboueljineane, L., & Frichi, Y. (2022)	Inefficient resource planning and coordination in emergency medical systems leading to high response times and poor coverage.	Discrete Event Simulation (DES) combined with simulation optimization for ambulance relocation and resource coordination.	Simulated EMS data (synthetic scenarios) representing emergency calls, ambulance resources and demand.	Optimized deployment and dynamic relocation significantly reduced average response time and improved system coverage compared to static policies.	Based on simulated data; results depend on accuracy of assumptions and input parameters.	Needs real-time data integration and ML-based prediction of events to support proactive resource allocation.
2	Anthony, T., Mishra, A. K., Stassen, W., & Son, J. (2021)	Misclassification of emergency calls in dispatch centres leading to inappropriate resource allocation.	Machine Learning: SVM, Random Forest, Logistic Regression, k-Nearest Neighbors; compared for call classification.	Real EMS call data from South African Emergency Dispatch Centres.	Machine learning models improved call classification accuracy; Random Forest and SVM performed best.	Single region data; class imbalance and variability in caller descriptions.	Generalization to other regions and integration with response time prediction models.
3	Butsingkorn, T., Apichottanakul, A., & Arunyanart, S. (2026)	Inaccurate forecasting of ambulance demand causing poor emergency response planning.	Integrated ML approach: Random Forest, SVR, XGBoost and ensemble methods.	Historical EMS demand data (Thailand) including time, location, weather and socioeconomic factors.	Ensemble model achieved highest accuracy (lower MAE and RMSE) compared to individual models in demand forecasting.	Limited to one country/region; external factors (e.g., traffic) not fully considered.	Include real-time traffic and operational data for more accurate short-term prediction.
4	Viglino, D., Vesin, A., Ruckly, S. et al. (2017)	Need to predict daily volume of emergency calls for better staffing and resource planning.	Generalized Additive Model (GAM) using calendar and epidemiological variables.	SAMU Emergency Call Center, Grenoble, France. Period: 2005–2011 (1,296,153 calls).	Model explained 89% of variance; predicted daily calls within ±100 calls for 90.5% of days.	Predicts only daily volume, not incident level or response time; single center study.	Extend to multiple regions and predict incident-level demand and response time.
5	Garg, T., Toshniwal, D., & Parida, M. (2026)	Variability and lack of interpretability in EMS demand prediction.	Meta-learning ensemble framework combining multiple ML models for robust and interpretable prediction.	Historical EMS demand data (multiple years; location not specified in abstract).	Proposed ensemble outperformed individual models with higher accuracy and better stability and interpretability.	Requires large quality dataset; computationally intensive.	Apply to real-time prediction and integrate with dispatch optimization systems.
6	Zaheeruddin, & Gupta, H. (2022)	Suboptimal allocation of ambulances among base stations leading to high response times.	Mutation-based Shuffled Frog Leaping Algorithm (mSFLA) compared with GA and PSO.	Simulated EMS environment: 50 ambulances, 11 base stations, Southern Delhi, India.	mSFLA reduced average response time by 12.23% compared to GA and PSO; faster convergence and stable solutions.	Simulation-based; real-world validation required.	Combine with ML-based demand prediction for dynamic ambulance allocation.
7	Hill, P. L. (2025)	Understanding and predicting factors affecting EMS response times.	Machine Learning models to identify key factors and predict response time.	EMS operational data (New Zealand; Wellington region, 2018–2022).	ML models predicted response time with good accuracy; traffic, distance and call type were key factors.	Regional dataset; external validity may be limited.	Develop generalizable model using multi-region data and real-time inputs.
8	M.F. Alshehri, J. P. (2020)	Over-triage and mis-triage in EMS call handling and dispatch.	Mixed Methods: quantitative analysis of call data + qualitative interviews.	Cape Town EMS (South Africa). Call data + interviews (2017–2018).	Identified causes of over-triage and provided recommendations to improve dispatch accuracy.	Small sample size; context-specific.	Use ML models for triage prediction and test in multiple settings.
9	Naess, L. E., Krüger, A. J., Uleberg, O., et al. (2024)	Busy ambulances increase response times and reduce system availability.	Machine Learning classification models to detect busy periods and analyze impact.	Norwegian EMS data (2016–2021; multiple municipalities).	Busy ambulance periods significantly increase response times; ML models effectively identified these periods.	Retrospective study; causality not fully established.	Predict future busy periods using advanced time-series and real-time data.
10	Neira-Rodado, D., Paz-Roa, J. C., Escobar, J. W., & Ortiz-Barrios, M. A. (2025)	Need for accurate spatiotemporal prediction of EMS events.	PCA + Clustering (K-means) + XGBoost regression for spatiotemporal forecasting.	EMS event data, Barranquilla, Colombia (2015–2022).	Proposed model achieved high accuracy in predicting location and time of future EMS events.	Limited to one city; dependent on historical data patterns.	Extend to multiple cities and integrate with resource allocation and routing.
11	Selvan, C., Anwar, B. H., Naveen, S., & et al. (2025)	Non-optimal ambulance routing causing delays in reaching patients.	Deep Neural Network (DNN) for ambulance route optimization in mobile dispatch system.	Road network and traffic data with ambulance dispatch records (India; location not specified).	DNN model found faster routes compared to traditional methods; reduced travel time and response delay.	Depends on quality of real-time traffic data.	Integrate with dynamic dispatch and real-time incident prediction systems.
12	Shahidian, N., Abreu, P., Santos, D., & Barbosa-Povoa, A. (2025)	Need for accurate short-term EMS demand forecasting.	Gradient Boosting, AdaBoost, Random Forest, SVR, LSTM; compared for short-term demand forecasting.	EMS call data (Portugal; period not specified).	Gradient Boosting and LSTM achieved best performance in short-term demand forecasting.	Single country data; external factors limited.	Include exogenous variables (e.g., traffic, weather) and test in other regions.
13	Wang, H., Ng, Q., Arulananandam, S., Tan, C., Ong, M. E., & Feng, M. (2023)	Need for accurate dispatch triage to prioritize ambulance allocation.	Machine Learning (e.g., XGBoost, Random Forest, Logistic Regression) for triage and prioritization.	Singapore EMS data (2018–2022; SCDF records).	ML-based triage model improved prioritization accuracy and could reduce response time.	Retrospective data; needs real-time implementation.	Deploy model in real-time dispatch systems and evaluate impact.
14	Yaseen, F. A., Alkhalidi, N. A., & Al-Raweshidy, H. S. (2022)	Lack of prioritization of emergency healthcare network traffic affecting QoS.	Machine Learning + Software Defined Networking (SDN) for traffic classification and priority management.	Simulated network traffic data (healthcare and emergency communication).	SHE framework improved traffic prioritization, reduced delay and enhanced QoS for emergency communications.	Simulation-based; real-world network validation needed.	Implement and test in live healthcare communication networks.
15	Yuan, Y. S. (2022)	Uncertainty in EMS system planning and resource allocation.	Wasserstein Distributionally Robust Chance Constrained Programming (DRCCP).	Synthetic/empirical emergency demand data for numerical experiments.	DRCCP approach provided robust solutions under uncertainty and outperformed traditional models.	Computationally complex; requires distributional assumptions.	Apply to large-scale real-world EMS planning with real data.

3. References

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